

BOSTON airbnb LISTINGS ANALYSIS



1. INTRODUCTION

2. DATA CARD

3. DATA CLEANING

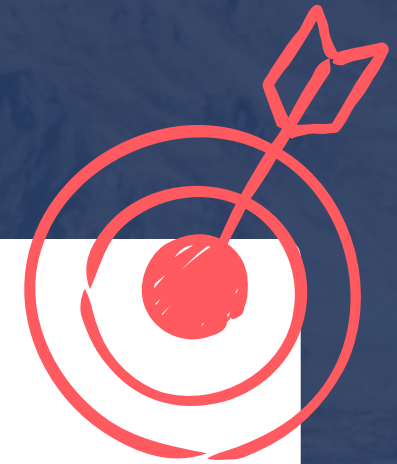
4. DESCRIPTIVE ANALYSIS

5. MULTIPLE LINEAR REGRESSION (**MLR**)

6. GENERALIZED LINEAR MODEL (**GLM**)

7. LIMITATION

8. CONCLUSION



Project Goal:

Identify determinants of  **airbnb** listing competitiveness.

Dependent Variable (Y)

Number of Reviews per month (proxy for listing competitiveness).

To make the data digestible, we grouped our 95 variables into these categories:



PRICING: Price



LISTING ATTRIBUTES: Room type, bed count, bed rooms count, property type, bathroom count, ACCOMMODATES, amenities count



LOCATION: Neighborhood, longitude/latitude



HOST PROFESSIONALISM: Response rate, Superhost status, acceptance rate, response time, host LISTING counts, identity verification.

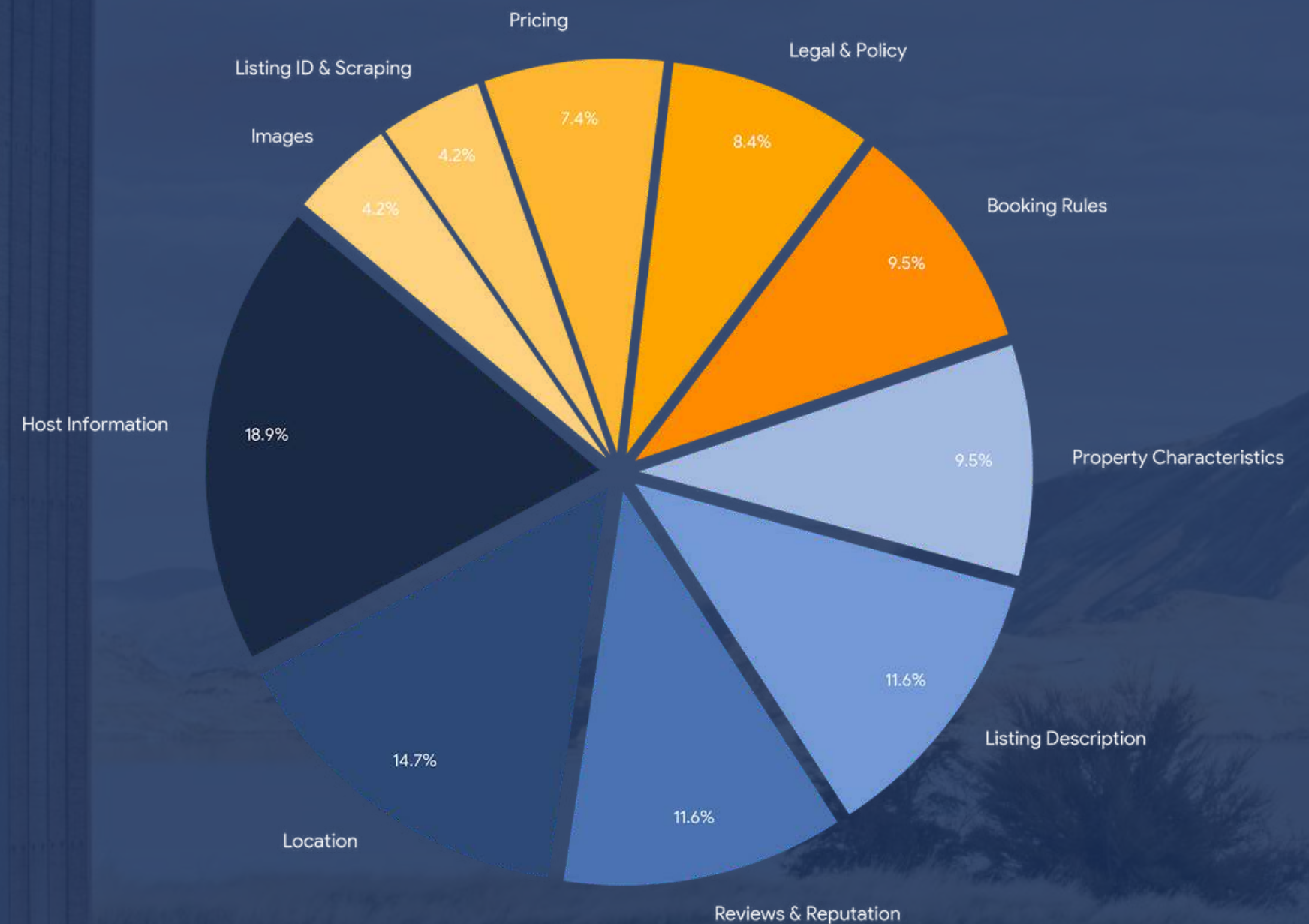


BOOKING RULES: Minimum stay, cancellation policy, instant booking, extra people, guest included.

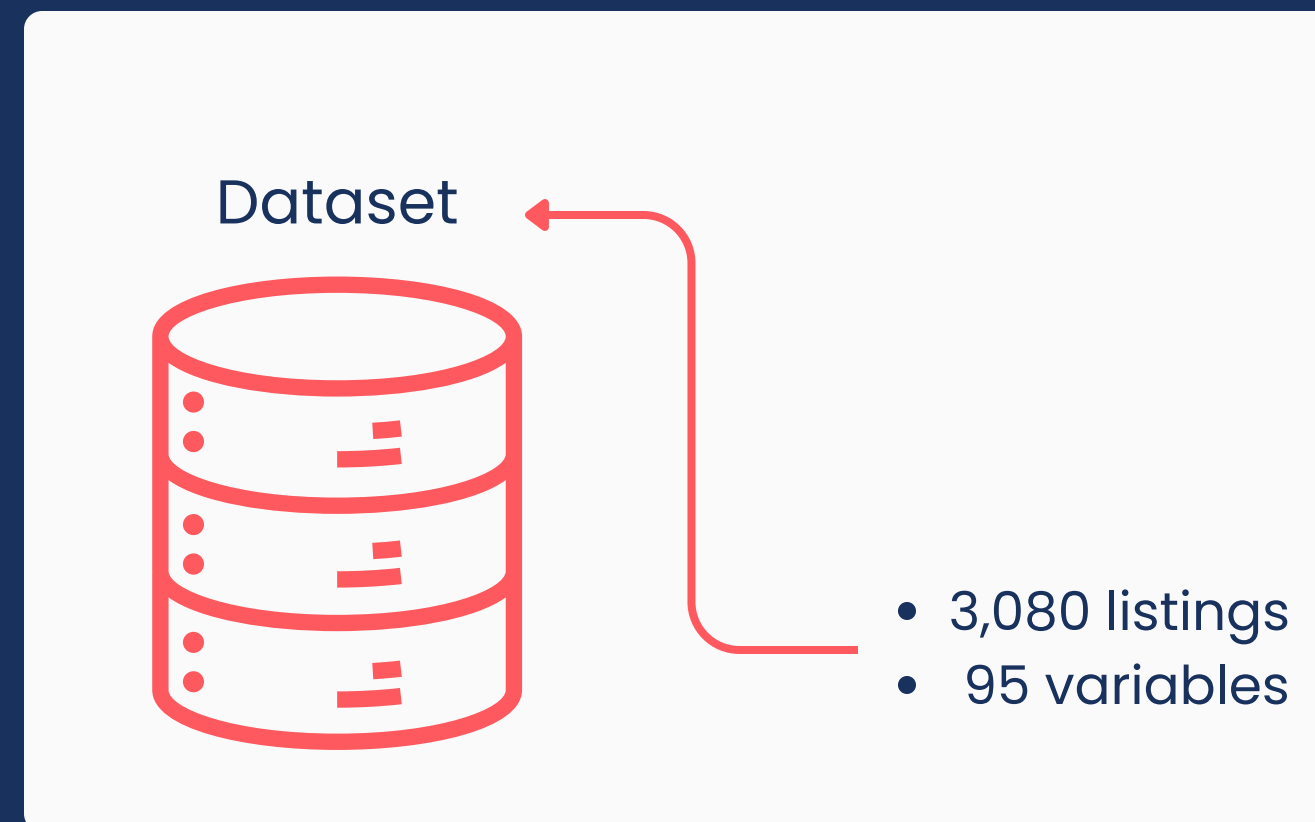
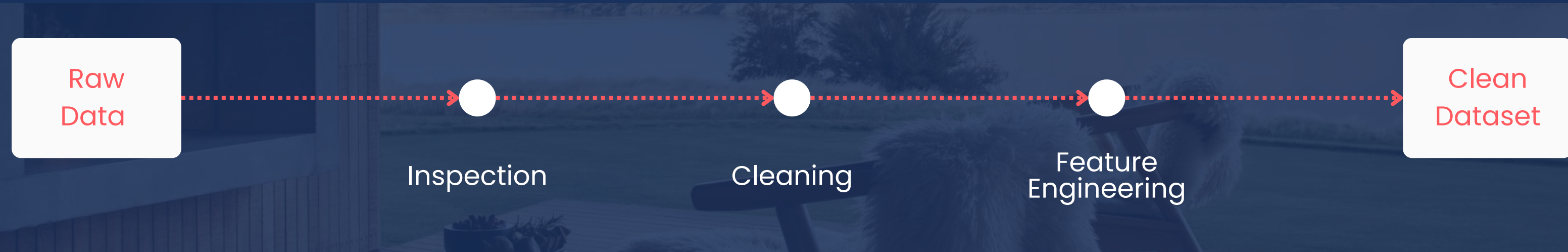


REPUTATION: Review scores

Distribution of 95 Original Variables by Category



DATA PREPARATION OVERVIEW



Data Inspection

Descriptive text / metadata



removed

Variables with 98–100% missing values



removed

Review score variables and reviews_per_month
contain missing values (~20–23%)



observations without ratings dropped

No duplicate observations detected

variables converted

Monetary variables (*price, extra_people*) removing currency symbols and converted to numeric.

Percentage variables (*host_response_rate, host_acceptance_rate*) converted to numeric

Binary variables (*host_is_superhost, host_identity_verified, instant_bookable*) were converted to 0/1 indicators.

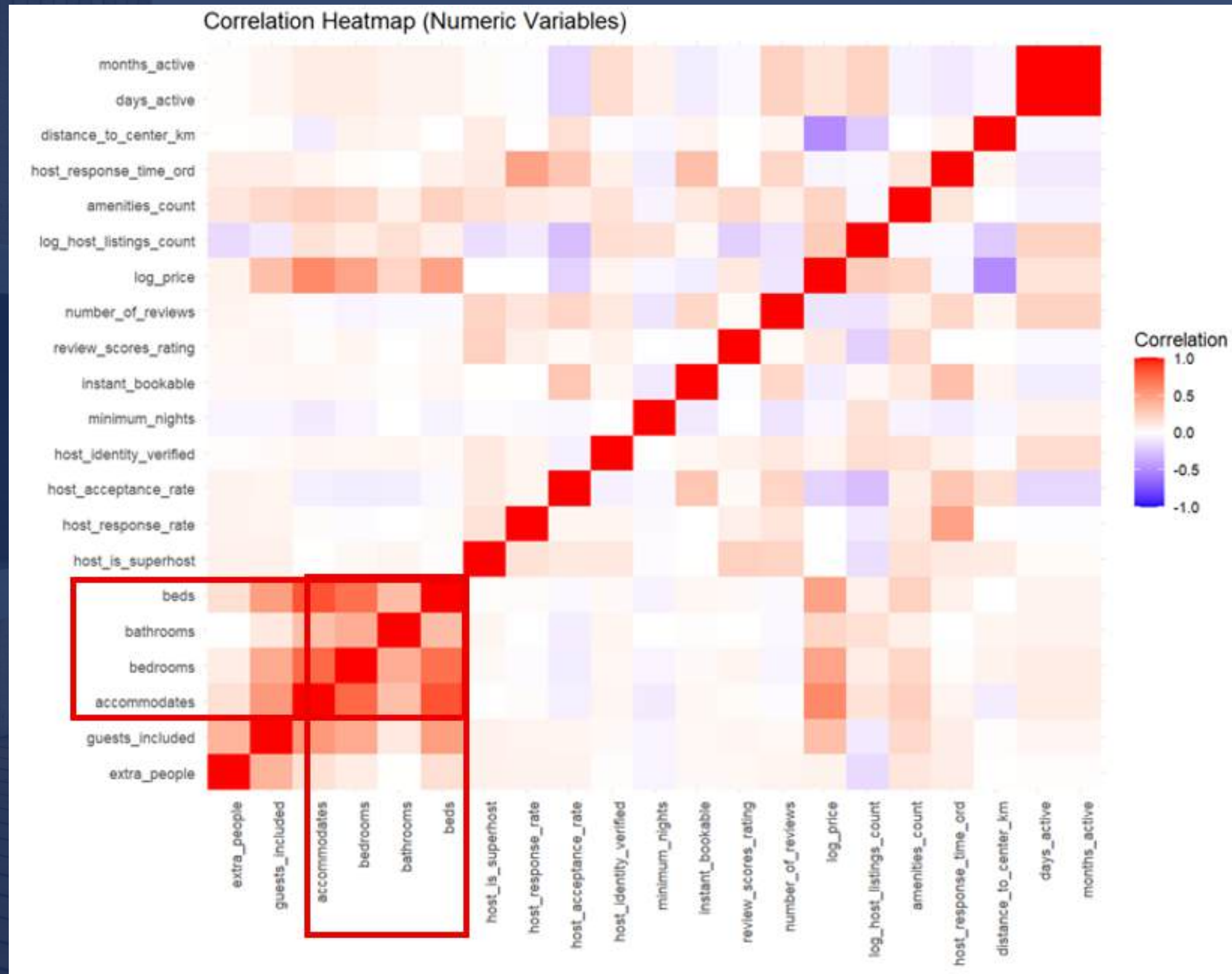
Date variables (*host_since, last_scraped*) were converted to Date format.

FEATURE ENGINEERING

Raw Variable	Transformation	Econometric Objectives
Location (lat/long)	distance_to_center	Capture spatial accessibility
Amenities (text)	amenities_count	Quantify property quality
Reviews/month	months_active	Exposure variable, using host since - last scraped
Price	log(price)	Elasticity interpretation
Host listings	log(1+count)	Diminishing marginal effects
Minimum nights	Capped at 30	Reduce outlier influence
Host response time	Ordinal scale (1-4)	Ordered professionalism proxy

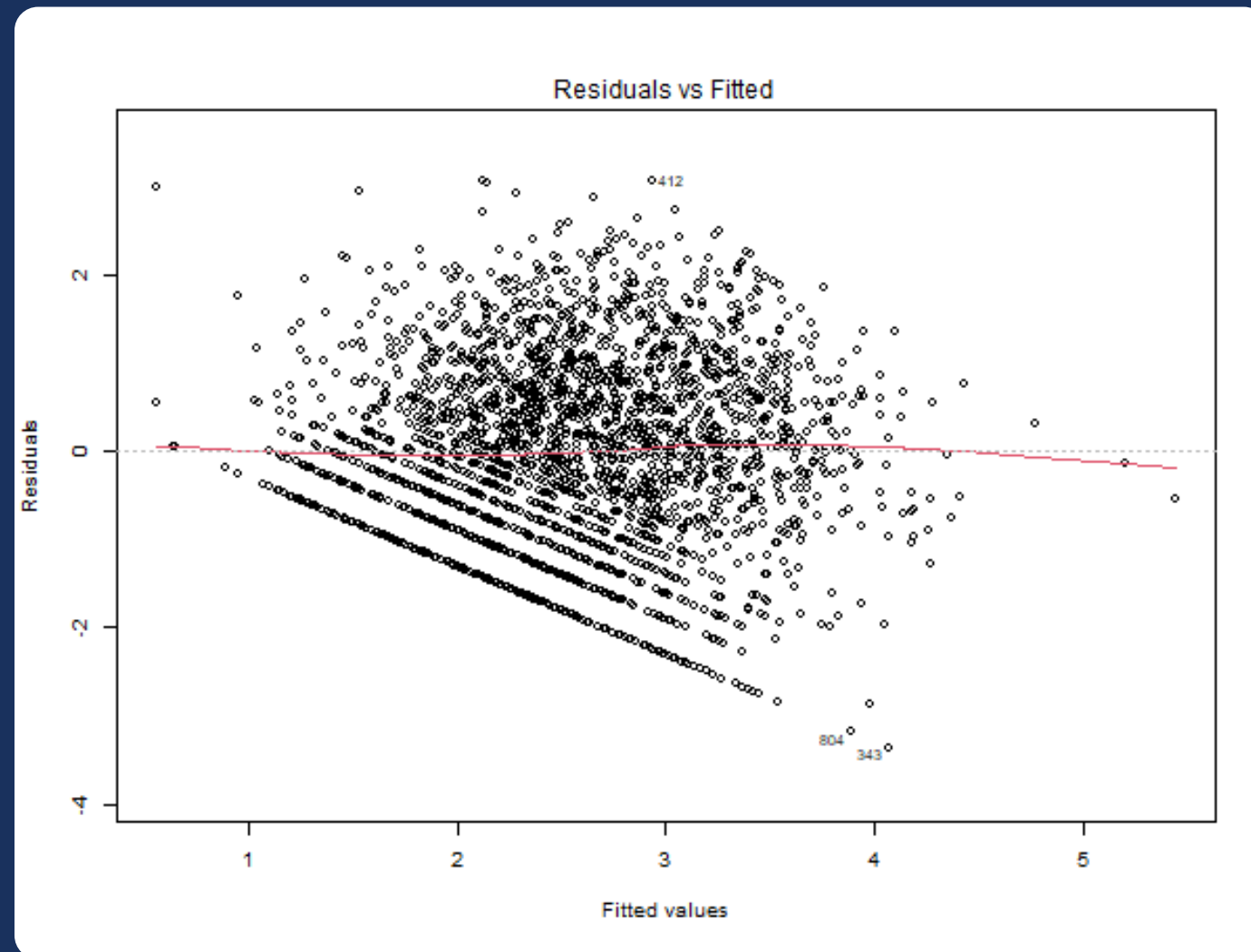


DESCRIPTIVE ANALYSIS



MULTIPLE LINEAR REGRESSION (MLR)

We tried using MLR, but...



KEY TAKEAWAYS:

MLR assumptions are violated
Results suggest transition to a count-based GLM models

Also, we are trying to see if our variables are justified, so:

Kitchen Sink	Simplified
Complete Predictors	Core Drivers Only

Homoscedasticity is heavily violated

Also:

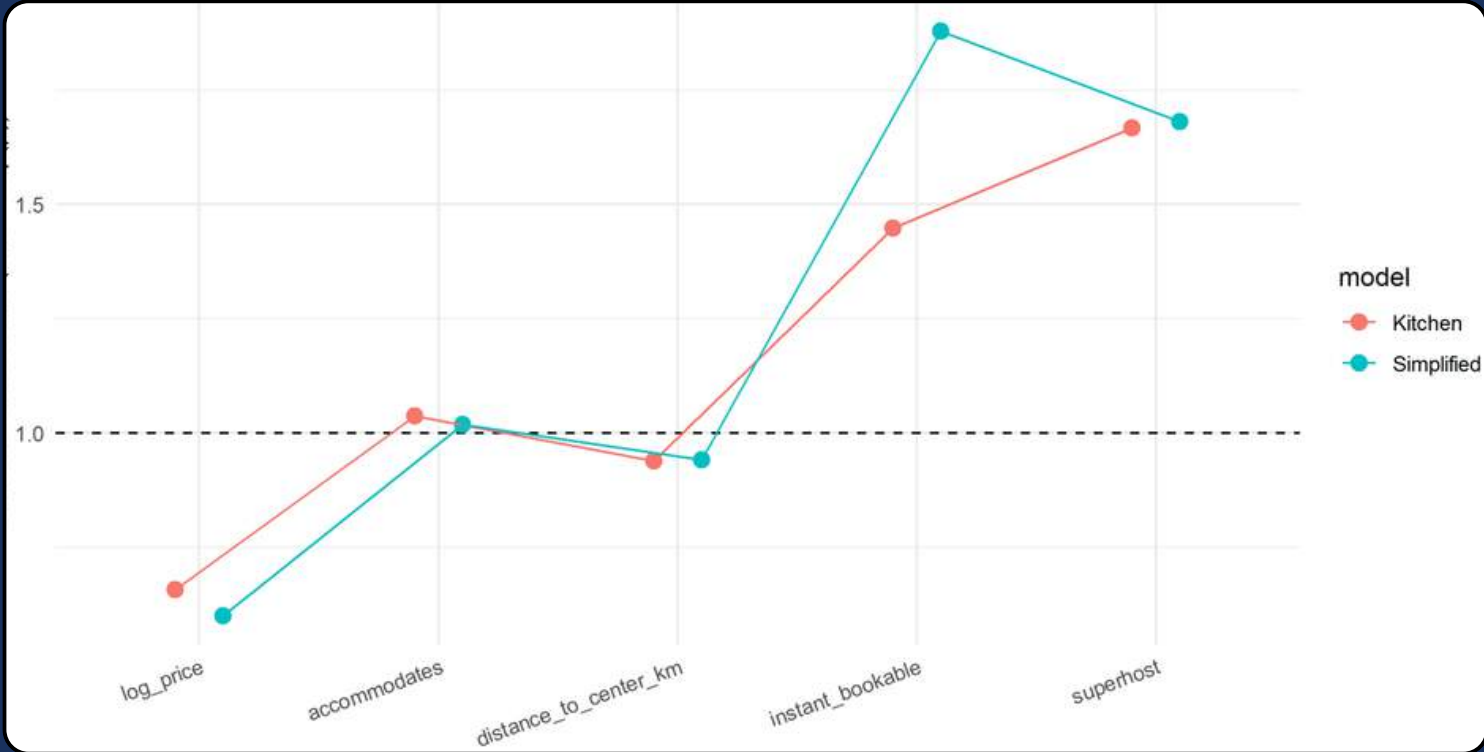
- Y variable has count characteristics
- YX relationship shouldn't be additive

...for the next model

MODELS COMPARED

Comparison	Kitchen Sink	Simplified
N	2513	
LRT (p-value)	<2.2e-1	
AIC	68,866.07	73,377.37

IRR Comparison Graph of Key Variables



The numbers favor the kitchen sink, but....

PEARSON DISPERSION

Kitchen Sink	30.4
Simplified	34.1

KEY TAKEAWAYS:

Poisson variance assumption is severely violated.
Results suggest transition to Negative Binomial model

...PROBLEM SOLVED

Comparison	Poisson	Negative Binomial
Dispersion	~30	~1
AIC	68,866.07	20,375.7

NB allows for overdispersion and provides a **better fit**

ANOTHER PROBLEM

Accomodates	+0.108
Beds	-0.079
Bedrooms	-0.051
Bathrooms	+0.039

beds and bedrooms produced counter-intuitive signs once controlling for accomodates

VARIABLE SOLUTION

Drop

- Beds
- Bedrooms
- Bathrooms

Keep

- Accomodates

Accomodates as primary size/capacity indicators

we think it's because:

Beds

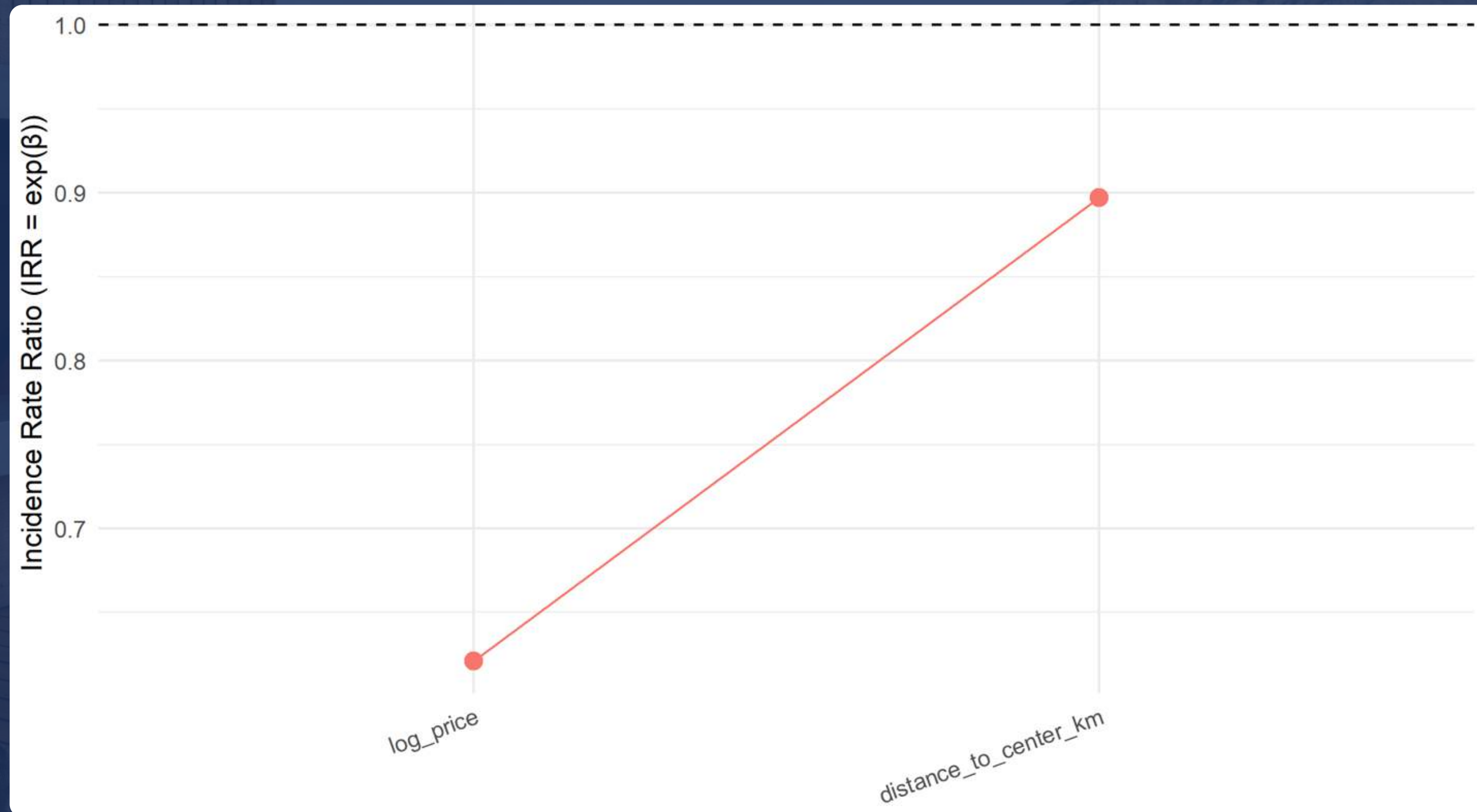
- Reduce privacy
- Reflect cheap configuration

Bedrooms

- Too much segmented layouts
- Reduce shared/common space

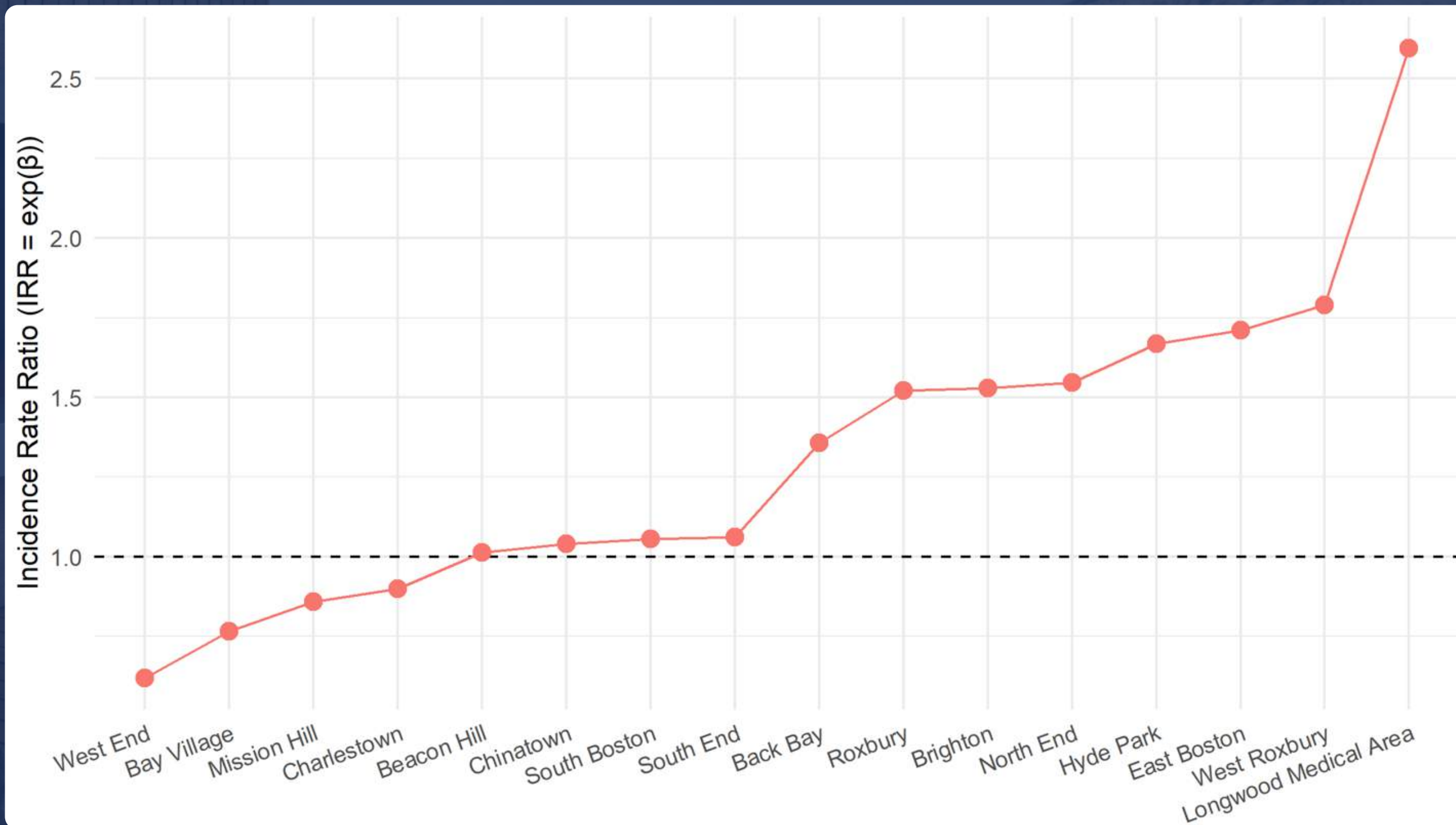
It does not means that more beds or bedrooms are worse

RESULT 1. PRICE AND LOCATION(A)

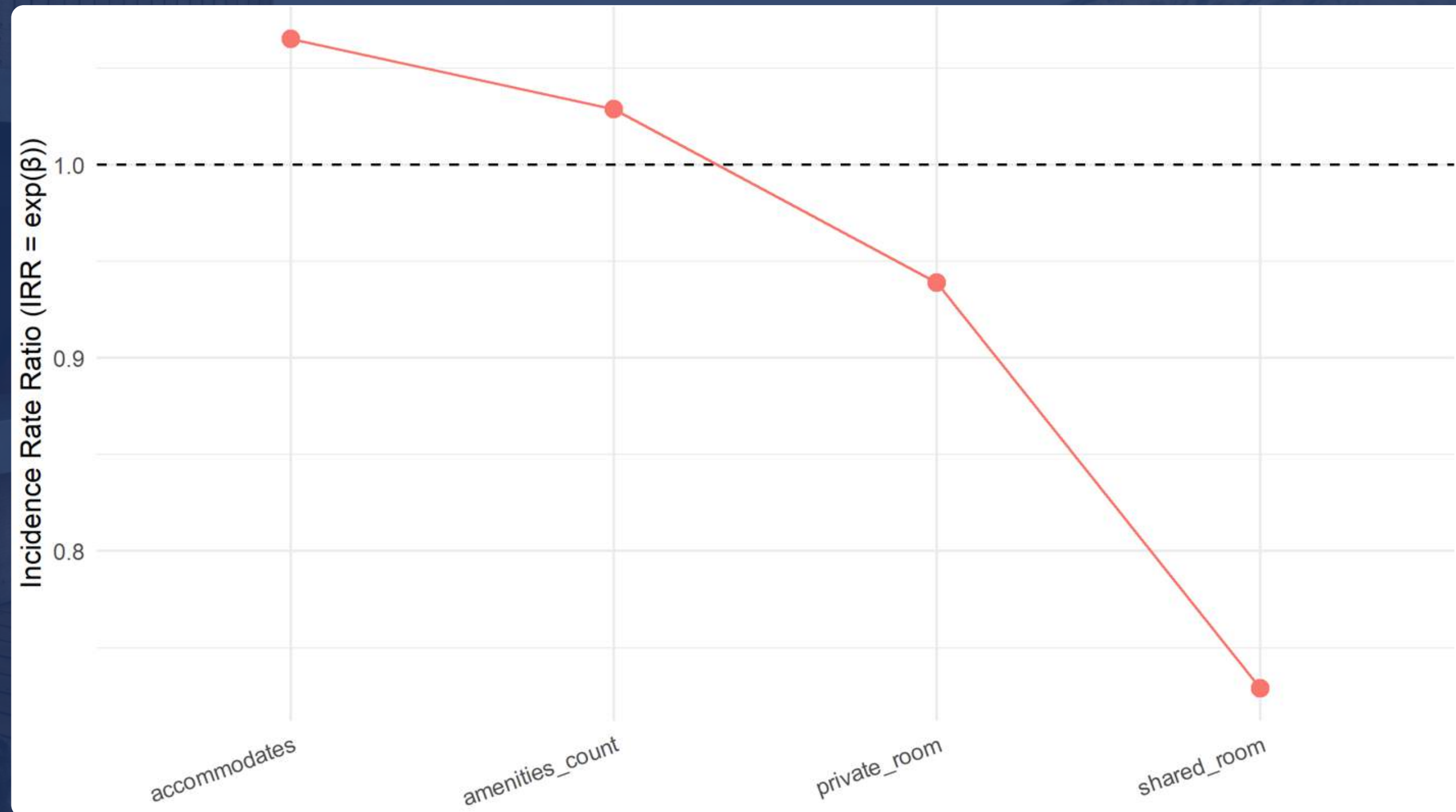


RESULT

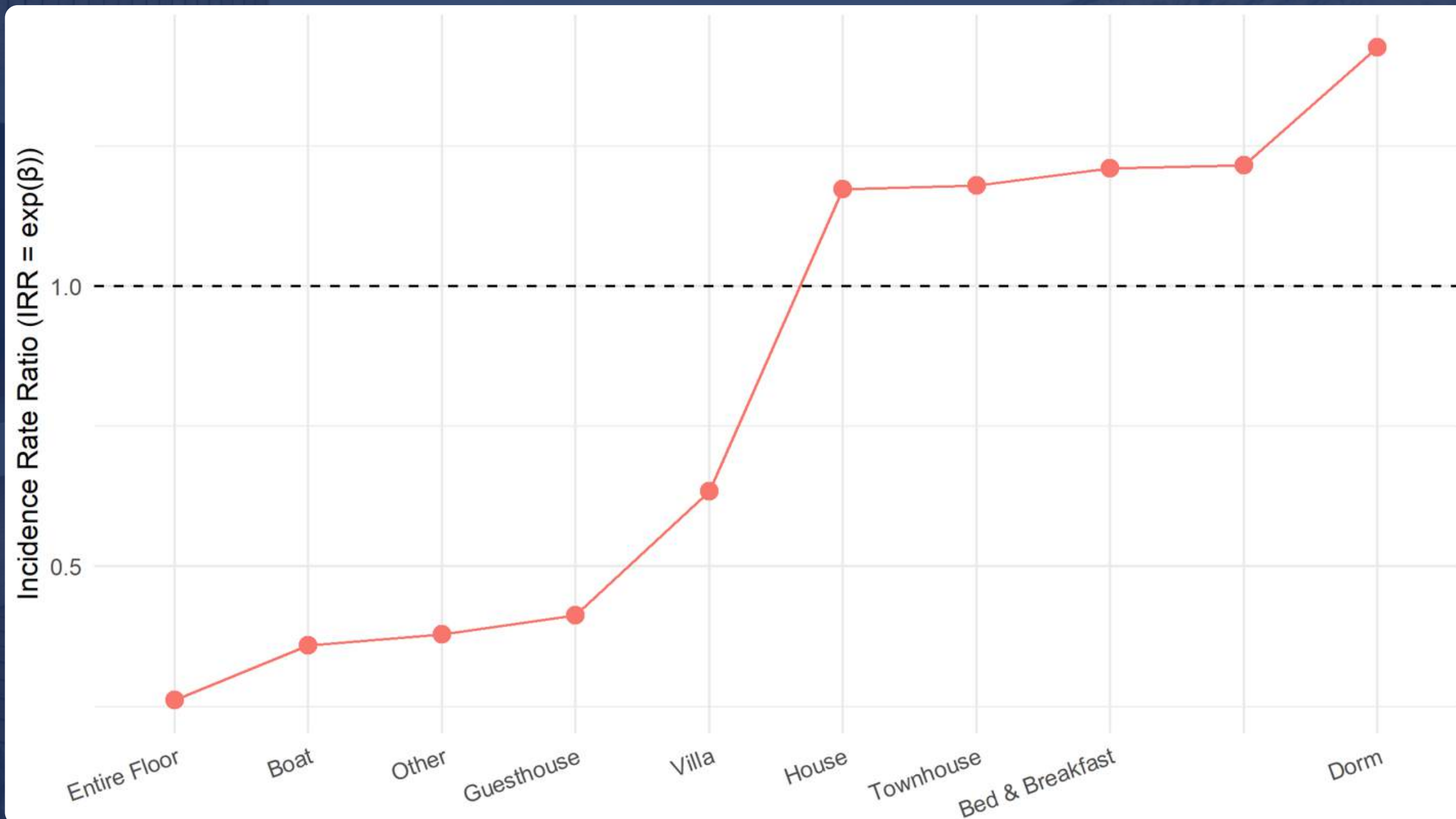
1. PRICE AND LOCATION(B) – NEIGHBORHOOD



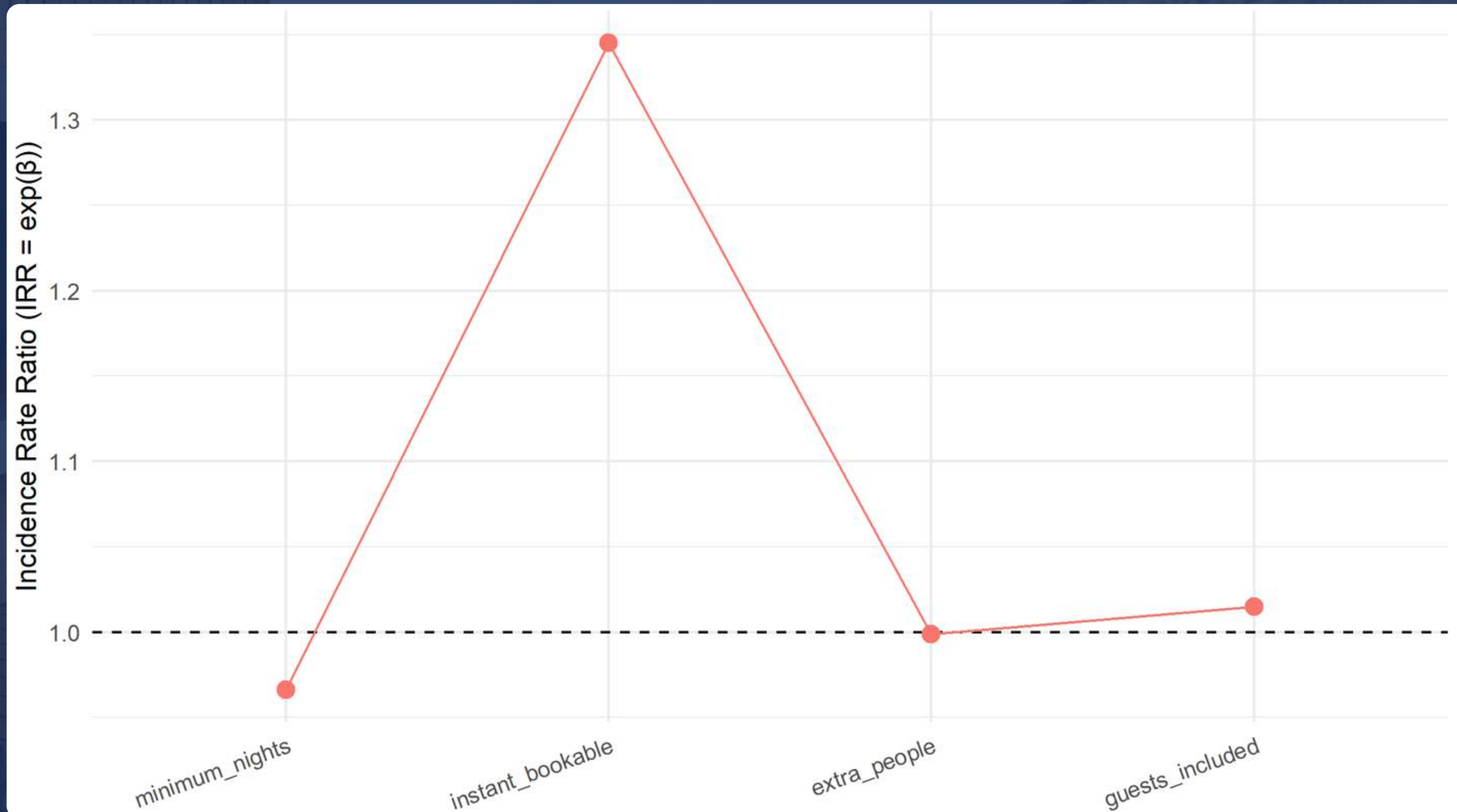
RESULT 2. LISTINGS ATTRIBUTES(A) – CORE ATTRIBUTES



RESULT 2. LISTINGS ATTRIBUTES(B) – PROPERTY TYPE

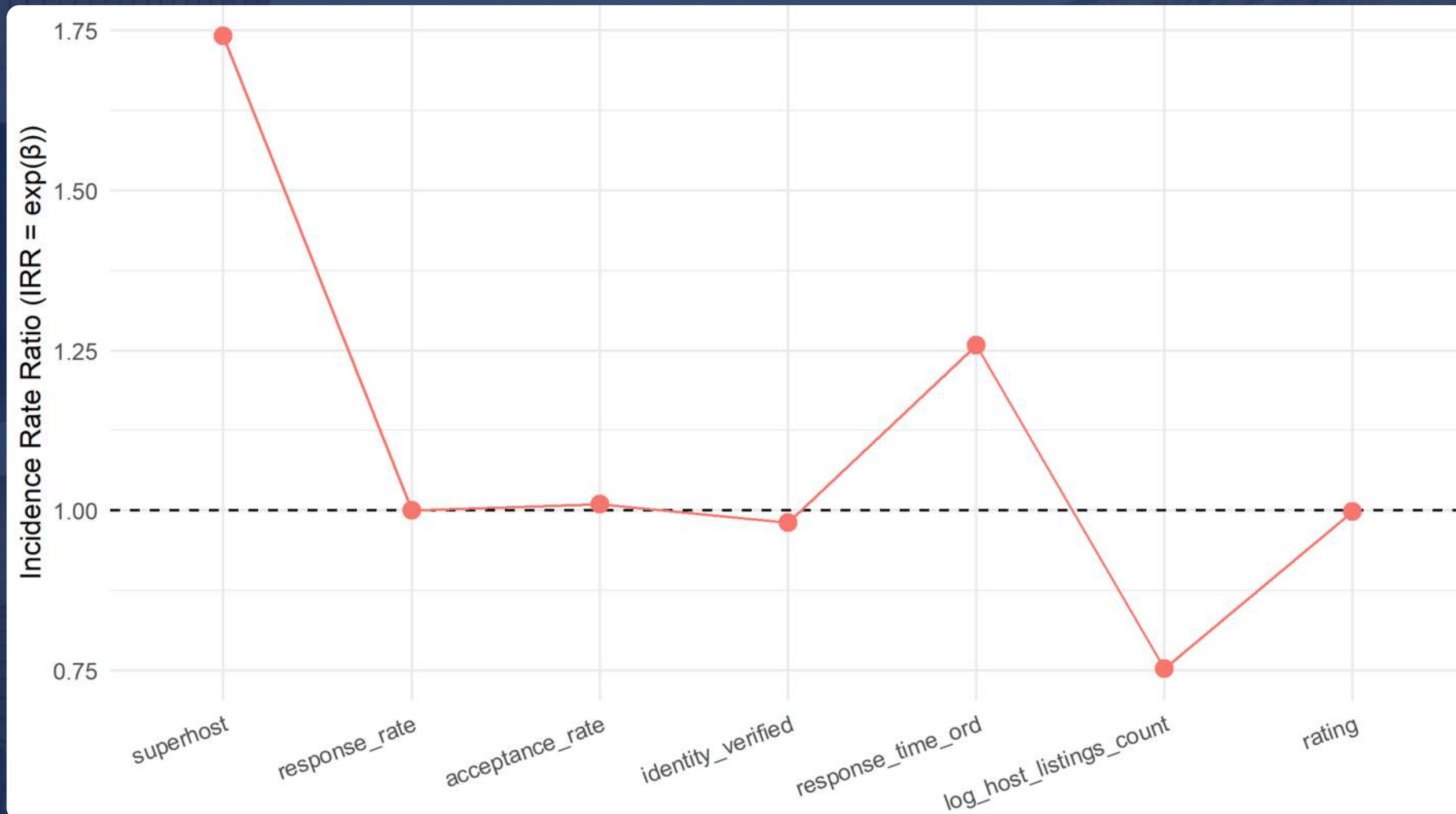


RESULT 3. BOOKING RULE



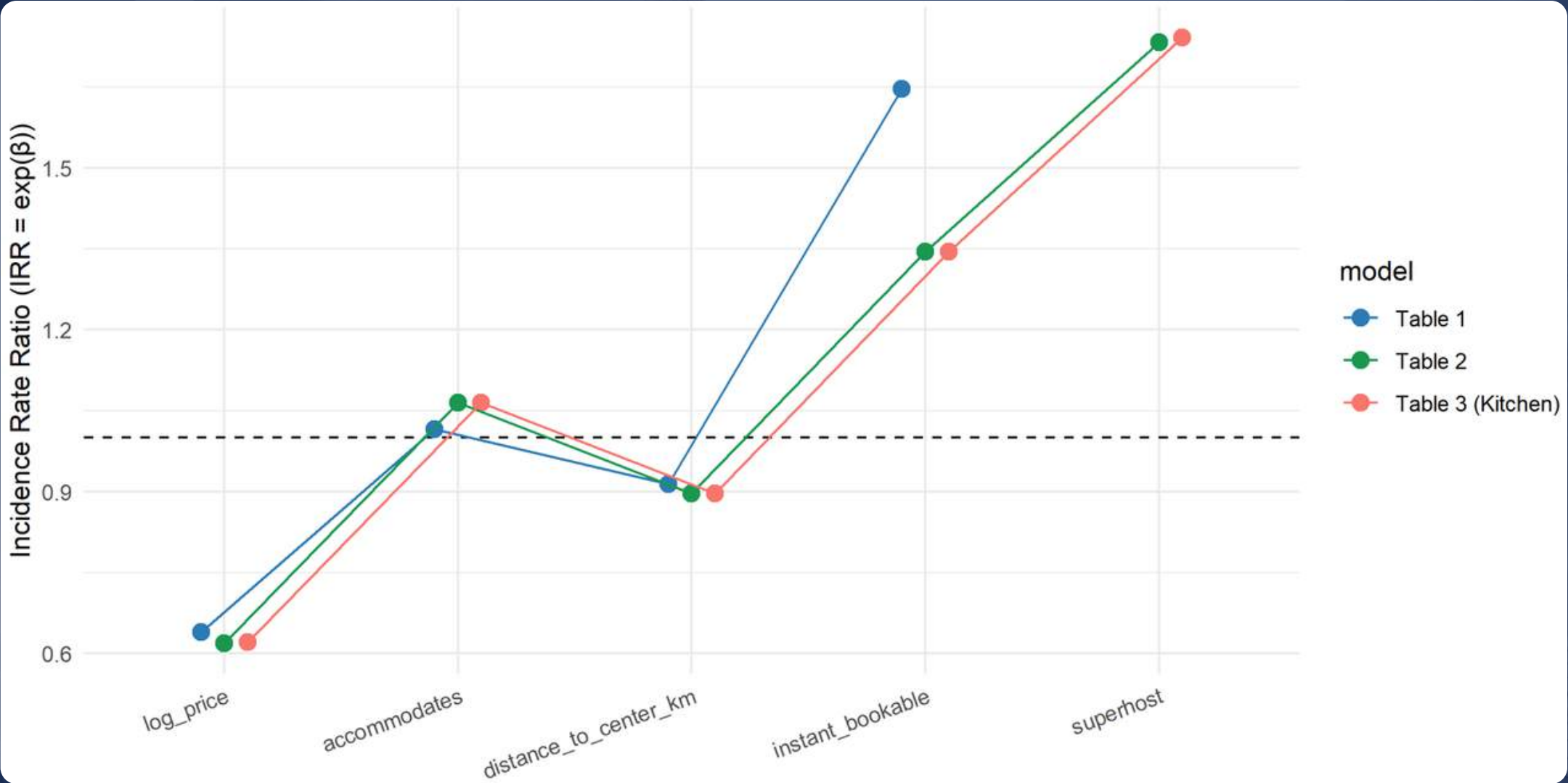
RESULT

4. HOST PROFESSIONALISM + REPUTATION



LADDER STABILITY

IRR Comparison



Other Comparison

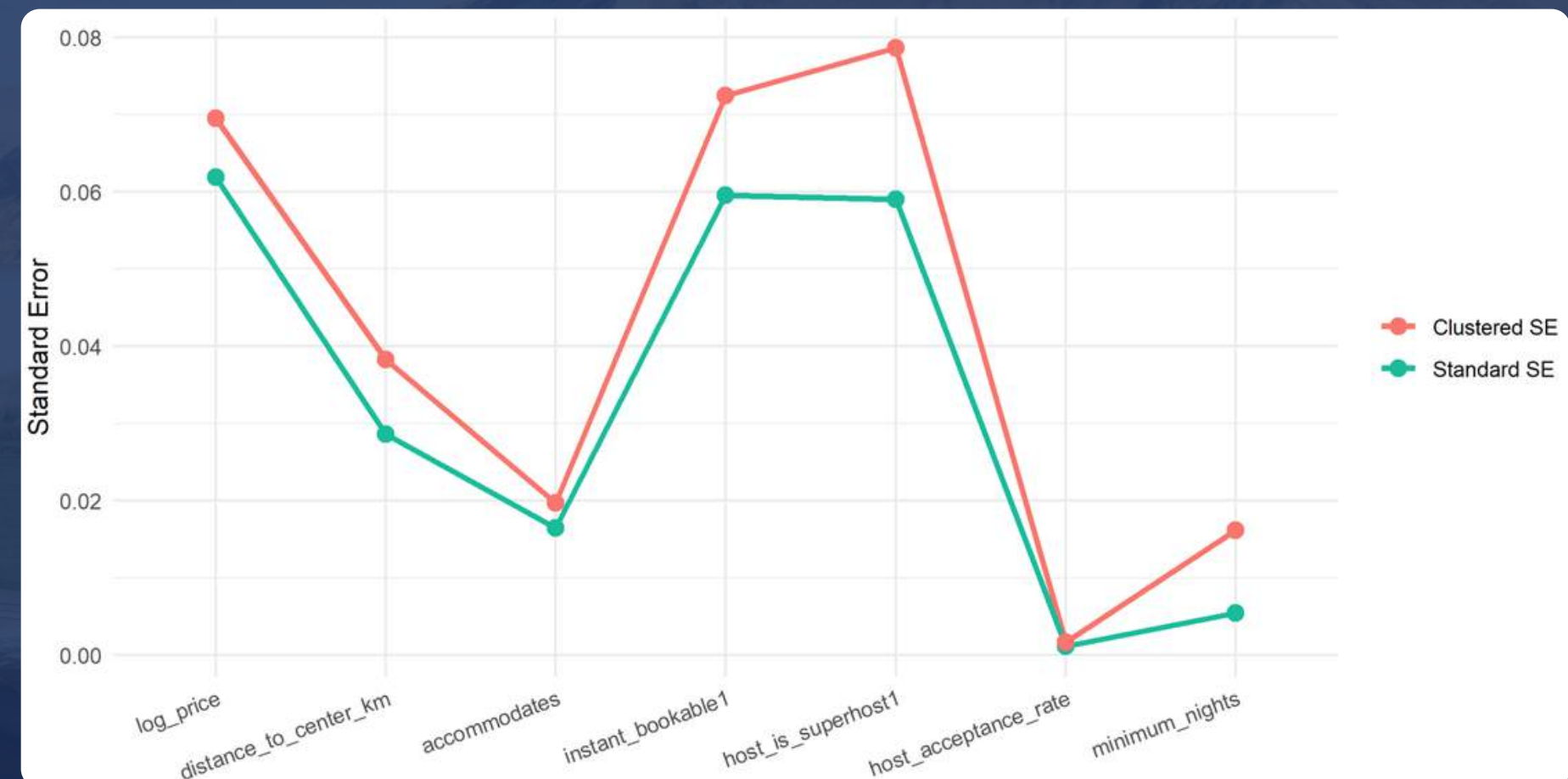
Variables	Table 1	Table 2	Table 3
θ (Theta)	0.897	1.077	1.077
AIC	20911	20374	20376
LRT (p-value)	~0		
		0.709	

ADDITIONALS: HOST CLUSTER

Multiple listings belong to the same host and may share unobserved characteristics (pricing strategy, management quality, branding).

Variables	p-Value
log_price	<0.001
distance	0.004
instant_bookable	<0.001
superhost	<0.001
acceptance_rate	<0.001
minimum_nights	0.033

INDEPENDENCE ASSUMPTIONS MAY BE VIOLATED



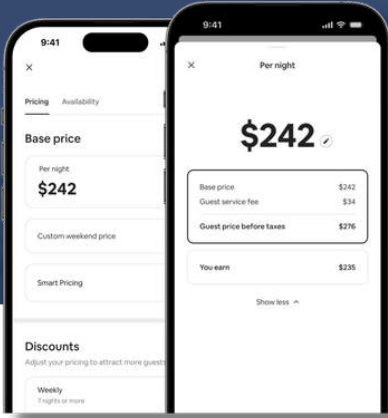
CONCLUSION

SE changed, but p-values are still acceptable (<0.05)
Main competitiveness drivers remain strong and statistically significant even after clustering.

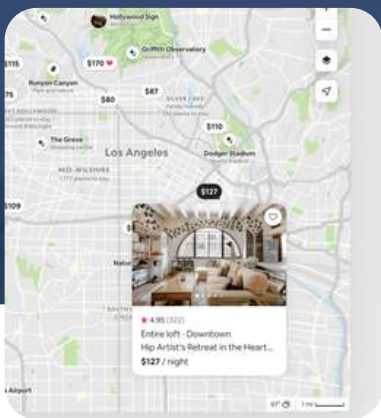
1. The analysis shows **relationships**, not causation
2. Potential **endogeneity** (pricing and host behavior)
3. **Host Cluster effect**
4. **Exposure proxy**
5. Model does not capture dynamic or **seasonal adjustments**
6. Reviews are used as a proxy for demand; **they do not directly measure bookings or revenue**

In future work, we could move closer to causal conclusions using better data and stronger methods

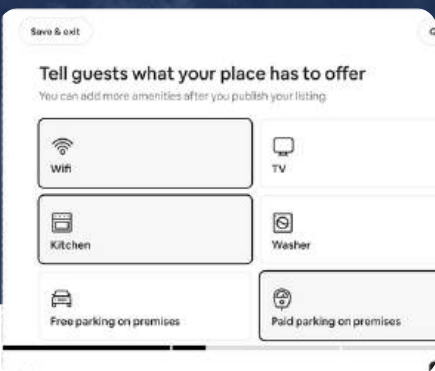
Airbnb is no longer a quality race, it's a competition beyond the baseline



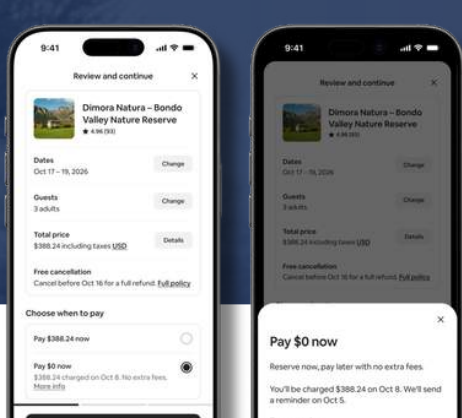
PRICING



LOCATION



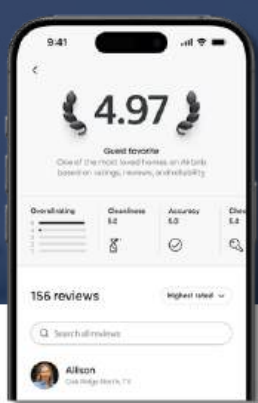
LISTINGS
ATTRIBUTE



BOOKING RULE



HOST
PROFESSIONALISM



REPUTATION

STRONG

Price

Distance to Center

Instant Bookable

Superhost Status

MODERATE

Amenities

Minimum Nights

Host Response Time,
Host Acceptance
Rate

WEAK

Guests included,
Extra people

Response rate,
Identity verification

Rating

KEY INSIGHTS:

1. Price has a strong negative effect → optimize pricing strategy to maximize demand
2. Central locations outperform → prioritize listings closer to city center
3. Instant booking significantly increases demand → reduce booking friction
4. **S**uperhost status and responsiveness increase trust → invest in host quality

SO WHAT SHOULD THE LANDLORDS DO?

1. Implement dynamic pricing strategies based on demand elasticity
2. Encourage instant booking adoption across listings
3. Improve host responsiveness through platform incentives
4. Highlight location advantage in listing positioning



BOSTON  **airbnb**
LISTINGS ANALYSIS

THANKYOU

for your attention